

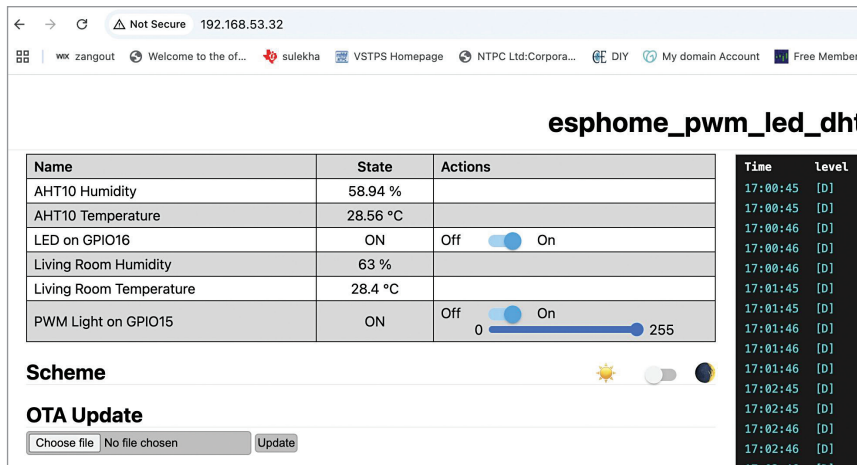


Fig. 6: Data on ESPHome page

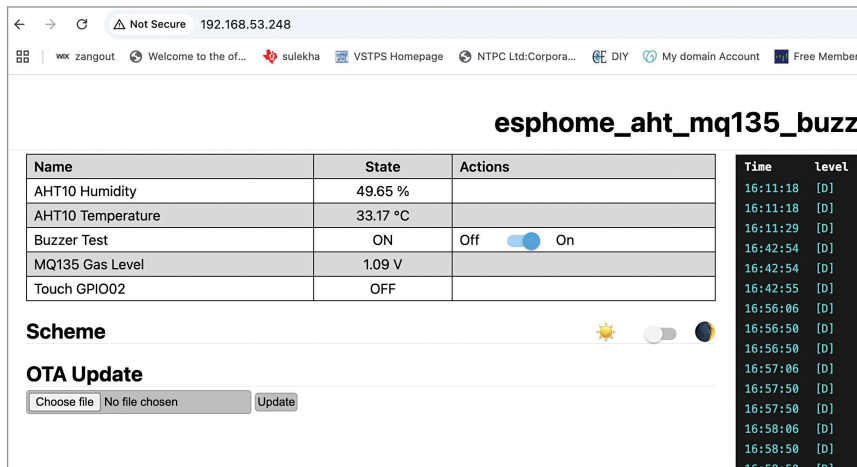


Fig. 7: MQ135 data on ESPHome

in Table 1. The author's prototype, built on a breadboard, is shown in Fig. 1, where the ESP32 controls LEDs through PWM.

Installation guide

Follow this installation guide to run the automation system using ESPHome on Windows, macOS, or Linux. ESPHome Automation is primarily a Python package installed via pip.

Prerequisites. Ensure Python (version 3.7 or higher) is installed on the system. Verify it using:

```
$ python --version
```

To install ESPHome via pip, open a terminal and run:

```
$ pip install esphome
```

Verify the installation:

```
$ esphome version
```

Version: 2024.10.3 (confirms ESPHome is installed.)

Optional: Install the ESPHome dashboard

(Note: As of the latest version, the ESPHome dashboard is included with ESPHome and does not require separate installation.)

```
$ pip install esphome-dashboard
```

Only if needed for older versions

Prepare a working directory. Create a directory where all ESPHome YAML files will reside:

```
$ mkdir /Users/somnathbera/esphome
```

```
$ cd /Users/somnathbera/esphome
```

The message appears on the

screen: 'Navigate to the directory.'

Create and compile the YAML configuration.

- To create a blank YAML file for your device:

```
$ esphome wizard my_device.yaml
```

- To compile and upload the YAML file to your device:

```
$ esphome run my_device.yaml
```

The message appears on the screen: 'Keep your device connected to the serial port'

Tip: The ESPHome dashboard can be used instead of the terminal for those who prefer a graphical interface.

Using ESPHome dashboard (GUI). To start the ESPHome Dashboard, run:

```
$ esphome dashboard /Users/somnathbera/esphome
```

This starts a local web server. Open a browser and navigate to:

```
http://localhost:6052
```

The ESPHome dashboard interface will display all available devices and YAML configurations for easy management. Fig. 2 shows the ESPHome dashboard.

Circuit diagram

Fig. 3 shows the circuit diagram for the ESPHome Automation System. It is built around the ESP32 (MOD1), AHT10, and DHT22 sensors, along with a few other components. A 5V regulated power supply is connected to the ESP32's 5V pin, while the 3.3V output from the ESP32 board powers the remaining circuit.

The schematic is straightforward. For any sensor, simply provide power, connect it to a GPIO pin, configure that GPIO in the *.yaml file, and install the script—it is that simple. In this prototype:

AHT10 sensor—SDA and SCL pins connected to G7 and G8 of ESP32.

DHT22 sensor—Data pin connected to G4 of ESP32.

The prototype using the DH22